

Kairui Hu

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EDUCATION

Imperial College London

MEng in Electrical and Electronic Engineering

Sep 2026 – Jun 2030 (Expected)

London, UK

EXPERIENCE

Research Assistant — Embodied AI & Robotics

Shanghai Innovation Institute

Aug 2026 – Present

Shanghai, China/Remote

- Initiated an independent research project on compute-efficient action selection in generative world models, combining multi-fidelity estimation with adaptive candidate elimination.
- Carried out the experimental work for CoPASS, spanning thermoelectric device characterisation, controlled-contact material recognition, and robotic grasping validation.

Autonomous Driving Algorithm Intern

Relaxing Technology

Jun 2026 – Aug 2026

Chongqing, China

- Refined lane-detection preprocessing using Gaussian smoothing, image binarisation, and Canny edge detection to suppress noise and extract lane-marking edges under challenging illumination.
- Curated and annotated challenging driving frames; analysed missed lane markings and spurious edges to guide preprocessing and post-processing adjustments.

PROJECTS

Multi-Fidelity Planning for World Models

Aug 2026 – Present

- Designed a two-stage planner that screens action sequences with low-cost predictions and uses confidence bounds to select promising candidates for higher-fidelity evaluation.
- Formulated a multilevel Monte Carlo estimator of relative action values, using coupled random paths across fidelity levels and shared randomness across candidates to reduce comparison variance.
- Proposed early termination of rollouts using remaining-return upper bounds, allocating further computation to unresolved comparisons most likely to change the selected action.

PUBLICATIONS

CoPASS: Active Thermotactile Sensing

Co-author

ICRA 2027, Under Review

- Extracted amplitude and normalised decay features from aligned, baseline-corrected thermal responses; evaluated shrinkage LDA with leave-one-block-out validation, correctly classifying 15/15 contacts across three material specimens.
- Conducted 15 robotic grasping trials using time-multiplexed Peltier excitation and Seebeck readout, validating repeatable thermal-response acquisition on a force-controlled gripper.

TECHNICAL SKILLS

Languages: Python, C++

Robotics & Machine Learning: ROS 2, PyTorch

Systems & Development: Linux, Git, CMake, GDB, unit testing

Scientific Computing & Computer Vision: NumPy, SciPy, pandas, OpenCV, Matplotlib

Experimental Methods: Sensor calibration, data acquisition, feature extraction, statistical classification

Others: Verilog, FPGA

HONOURS & ACADEMIC ACHIEVEMENTS

40th, 41st Chinese Physics Olympiad (CPhO)

First Prize

Sep 2023 & Sep 2024

Engineering and Science Admissions Test (ESAT)

Full Marks (9.0/9.0) in Math 1, 2 and Physics, Ranked 1st in the UK

Sep 2025

Physics Aptitude Test (PAT)

95.3/100, Top 20 in the UK

Sep 2025